Project 2

COMP-4250

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Question 1

J48 Results

J48 pruned tree

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lym_nodes_dimin <= 1

| changes_in_node = no

| | defect_in_node = no: normal (3.0/1.0)

| | defect_in_node = lacunar: malign_lymph (2.0)

| | defect_in_node = lac_margin: normal (0.0)

| | defect_in_node = lac_central: normal (0.0)

| changes_in_node = lacunar

| | exclusion_of_no = no: metastases (10.0/1.0)

| | exclusion_of_no = yes

| | | special_forms = no: metastases (3.0/1.0)

| | | special_forms = chalices

| | | | lym_nodes_enlar <= 2: malign_lymph (3.0)

| | | | lym_nodes_enlar > 2: metastases (2.0)

| | | special_forms = vesicles: malign_lymph (19.0/1.0)

| changes_in_node = lac_margin

| | block_of_affere = no

| | | extravasates = no

| | | | lymphatics = normal: metastases (0.0)

| | | | lymphatics = arched

| | | | | early_uptake_in = no: metastases (5.0/1.0)

| | | | | early_uptake_in = yes: malign_lymph (4.0/1.0)

| | | | lymphatics = deformed: metastases (5.0)

| | | | lymphatics = displaced: malign_lymph (1.0)

| | | extravasates = yes: malign_lymph (4.0)

| | block_of_affere = yes: metastases (56.0/3.0)

| changes_in_node = lac_central

| | no_of_nodes_in <= 1

| | | block_of_affere = no: malign_lymph (3.0)

| | | block_of_affere = yes: metastases (2.0)

| | no_of_nodes_in > 1: malign_lymph (20.0)

lym_nodes_dimin > 1

| by_pass = no: metastases (2.0/1.0)

| by_pass = yes: fibrosis (4.0)

Number of Leaves : 21

Size of the tree : 34

Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	114	77.027 %
Incorrectly Classified Instances	34	22.973 %
Kappa statistic	0.5736	
Mean absolute error	0.1304	
Root mean squared error	0.3151	
Relative absolute error	48.619 %	
Root relative squared error	86.5138 %	
Total Number of Instances	148	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.014	0.500	1.000	0.667	0.702	0.991	0.500	normal
	0.790	0.194	0.831	0.790	0.810	0.594	0.788	0.737	metastases
	0.754	0.195	0.730	0.754	0.742	0.556	0.777	0.718	malign_lymph
	0.500	0.014	0.500	0.500	0.500	0.486	0.744	0.389	fibrosis
Weighted Avg.	0.770	0.187	0.776	0.770	0.772	0.577	0.785	0.717	

=== Confusion Matrix ===

a	b	c	d	<-- classified as
2	0	0	0	a = normal

 1 64 15 1 | b = metastases

 1 13 46 1 | c = malign_lymph

 0 0 2 2 | d = fibrosis

JRIP Results

JRIP rules:

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(lymphatics = normal) => class=normal (2.0/0.0)

(lym_nodes_dimin >= 2) and (by_pass = yes) => class=fibrosis (4.0/0.0)

(no_of_nodes_in >= 3) and (special_forms = vesicles) => class=malign_lymph (41.0/5.0)

(block_of_affere = no) and (extravasates = yes) => class=malign_lymph (8.0/0.0)

(changes_in_node = lac_central) => class=malign_lymph (8.0/2.0)

 => class=metastases (85.0/11.0)

Number of Rules : 6

Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances 115 77.7027 %

Incorrectly Classified Instances 33 22.2973 %

Kappa statistic 0.5725

Mean absolute error	0.1414
Root mean squared error	0.3108
Relative absolute error	52.7427 %
Root relative squared error	85.3428 %
Total Number of Instances	148

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.000	0.000	?	0.000	?	?	0.687	0.038	normal
	0.827	0.254	0.798	0.827	0.812	0.576	0.805	0.808	metastases
	0.738	0.172	0.750	0.738	0.744	0.567	0.780	0.715	malign_lymph
	0.750	0.007	0.750	0.750	0.750	0.743	0.872	0.694	fibrosis
Weighted Avg.	0.777	0.210	?	0.777	?	?	0.795	0.756	

=== Confusion Matrix ===

a	b	c	d	<-- classified as
0	1	1	0	a = normal
0	67	14	0	b = metastases
0	15	45	1	c = malign_lymph
0	1	0	3	d = fibrosis

Question 2

C4.5

=== Summary ===

Correctly Classified Instances	420	97.2222 %
Incorrectly Classified Instances	12	2.7778 %
Kappa statistic	0.9444	
Mean absolute error	0.0892	
Root mean squared error	0.1831	
Relative absolute error	17.8311 %	
Root relative squared error	36.5759 %	
Total Number of Instances	432	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.053	0.944	1.000	0.971	0.946	0.983	0.964	0
	0.947	0.000	1.000	0.947	0.973	0.946	0.983	0.981	1
Weighted Avg.	0.972	0.025	0.974	0.972	0.972	0.946	0.983	0.973	

=== Confusion Matrix ===

a	b	<-- classified as
204	0	a = 0
12	216	b = 1

RIPPER

=== Summary ===

Correctly Classified Instances	390	90.2778 %
Incorrectly Classified Instances	42	9.7222 %
Kappa statistic	0.8053	
Mean absolute error	0.1314	
Root mean squared error	0.277	
Relative absolute error	26.2643 %	
Root relative squared error	55.3461 %	
Total Number of Instances	432	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.912	0.105	0.886	0.912	0.899	0.806	0.938	0.879	0
	0.895	0.088	0.919	0.895	0.907	0.806	0.938	0.942	1
Weighted Avg.	0.903	0.096	0.903	0.903	0.903	0.806	0.938	0.912	

=== Confusion Matrix ===

a	b	<-- classified as
186	18	a = 0
24	204	b = 1

k-Nearest Neighbor

=== Summary ===

Correctly Classified Instances	378	87.5 %
Incorrectly Classified Instances	54	12.5 %
Kappa statistic	0.7512	
Mean absolute error	0.191	
Root mean squared error	0.3228	
Relative absolute error	38.1693 %	
Root relative squared error	64.5029 %	
Total Number of Instances	432	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.941	0.184	0.821	0.941	0.877	0.758	0.933	0.902	0
	0.816	0.059	0.939	0.816	0.873	0.758	0.933	0.935	1
Weighted Avg.	0.875	0.118	0.883	0.875	0.875	0.758	0.933	0.919	

=== Confusion Matrix ===

a	b	<-- classified as
192	12	a = 0
42	186	b = 1

Naïve Bayesian

=== Summary ===

Correctly Classified Instances	420	97.2222 %
Incorrectly Classified Instances	12	2.7778 %
Kappa statistic	0.9444	
Mean absolute error	0.1863	
Root mean squared error	0.2323	
Relative absolute error	37.2363 %	
Root relative squared error	46.4131 %	
Total Number of Instances	432	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.053	0.944	1.000	0.971	0.946	0.975	0.961	0
	0.947	0.000	1.000	0.947	0.973	0.946	0.975	0.985	1
Weighted Avg.	0.972	0.025	0.974	0.972	0.972	0.946	0.975	0.973	

=== Confusion Matrix ===

a	b	<-- classified as
204	0	a = 0
12	216	b = 1

Neural Network

=== Summary ===

Correctly Classified Instances	404	93.5185 %
Incorrectly Classified Instances	28	6.4815 %
Kappa statistic	0.8709	
Mean absolute error	0.068	
Root mean squared error	0.2322	
Relative absolute error	13.5875 %	
Root relative squared error	46.3993 %	
Total Number of Instances	432	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.123	0.879	1.000	0.936	0.878	0.967	0.941	0
	0.877	0.000	1.000	0.877	0.935	0.878	0.967	0.981	1
Weighted Avg.	0.935	0.058	0.943	0.935	0.935	0.878	0.967	0.962	

=== Confusion Matrix ===

a	b	<-- classified as
204	0	a = 0
28	200	b = 1

The overall best performance in terms of accuracy was C4.5 and Naïve Bayesian. This could be because there are only 2 classifications possible, so it being less complex makes more Naïve choices good.

Question 3

C4.5

=== Summary ===

Correctly Classified Instances	178	85.9903 %
Incorrectly Classified Instances	29	14.0097 %
Kappa statistic	0.7168	
Mean absolute error	0.1958	
Root mean squared error	0.3288	
Relative absolute error	39.4502 %	
Root relative squared error	65.6306 %	
Total Number of Instances	207	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.776	0.064	0.916	0.776	0.840	0.725	0.901	0.857	+
	0.936	0.224	0.823	0.936	0.876	0.725	0.901	0.872	-
Weighted Avg.	0.860	0.149	0.867	0.860	0.859	0.725	0.901	0.865	

=== Confusion Matrix ===

a b <-- classified as

76 22 | a = +

7 102 | b = -

Naïve Bayesian

=== Summary ===

Correctly Classified Instances	156	75.3623 %
Incorrectly Classified Instances	51	24.6377 %
Kappa statistic	0.4968	
Mean absolute error	0.2468	
Root mean squared error	0.4633	
Relative absolute error	49.7186 %	
Root relative squared error	92.494 %	
Total Number of Instances	207	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.561	0.073	0.873	0.561	0.683	0.529	0.880	0.869	+
	0.927	0.439	0.701	0.927	0.798	0.529	0.880	0.887	-
Weighted Avg.	0.754	0.266	0.783	0.754	0.744	0.529	0.880	0.879	

=== Confusion Matrix ===

a b <-- classified as

55 43 | a = +

8 101 | b = -

Neural Networks

=== Summary ===

Correctly Classified Instances	160	77.2947 %
Incorrectly Classified Instances	47	22.7053 %
Kappa statistic	0.5401	
Mean absolute error	0.2173	
Root mean squared error	0.4352	
Relative absolute error	43.7768 %	
Root relative squared error	86.8833 %	
Total Number of Instances	207	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.663	0.128	0.823	0.663	0.734	0.550	0.869	0.864	+
	0.872	0.337	0.742	0.872	0.802	0.550	0.869	0.840	-
Weighted Avg.	0.773	0.238	0.780	0.773	0.770	0.550	0.869	0.851	

=== Confusion Matrix ===

a b <-- classified as

65 33 | a = +

14 95 | b = -

Question 4

Accuracy Table (%)

Dataset	C4.5	Ripper	Naïve Bayesian	K-Nearest Neighbors	Neural Networks
Ecoli	84.2262	80.6548	85.4167	80.3571	86.0119
Glass	65.8879	69.6262	49.5327	70.5607	69.1589
Image	89.0476	79.5238	77.619	87.1429	88.5714

Runtime Table (seconds)

Dataset	C4.5	Ripper	Naïve Bayesian	K-Nearest Neighbors	Neural Networks
Ecoli	0.01	0.04	0	0	0.33
Glass	0	0.03	0	0	0.23
Image	0	0.02	0	0	0.51

C4.5, Ripper, Naïve Bayesian, and K-Nearest Neighbors look to have an inconsistency in at least one of the datasets used compared to the others. Ecoli has worse accuracy using Ripper and K-Nearest Neighbors, Glass is bad with C4.5 and Naïve Bayesian, and Image is bad with Ripper and Naïve Bayesian. Using neural networks looks to be the most consistent for any type of data.

Although neural networks have the best accuracy, they are the worst in terms of runtime. All the other classifications have near 0 runtimes. Neural networks are 20 times longer than the rest.